3.9 Inventing a New Function

WORKSHEET

The question

STEP II 2021, Question 4

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2021 STEP papers (a zip file):

www.ocr.org.uk/Images/696588-2021-past-papers.zip

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself What does it actually mean to "solve" an equation? Is $x = \sqrt{2}$ a solution of $x^2 = 2$, or just a name for the solution?

STAGE 1 The curve $y = xe^x$

Ask yourself What happens to xe^x as $x \to +\infty$ and as $x \to -\infty$? Where does the curve cross the axes?

STAGE 3 Four equations

Ask yourself For (b) and (c), the unknown appears as $x \ln x$. What substitution turns $x \ln x$ into something of the form ue^u?

Pause and try Now try (d), $x = 3 \ln x$. (*Hint:* exponentiate and aim for (something) $e^{\text{something}}$ with the something being a multiple of x.)

Try this next

1 **Two, four, and one more.** The equation $x^2 = 2^x$ has the obvious solutions $x = 2$ and $x = 4$ (compare Section 3.5). Show that it also has a negative solution, and that this solution is

$$x = -\frac{2}{\ln 2}\, \mathrm{g}\left(\frac{\ln 2}{2}\right) \approx -0.767.$$

2 **A derivative.** By differentiating $\mathrm{g}(x)e^{\mathrm{g}(x)} = x$ implicitly, show that, for $x \neq 0$,

$$\mathrm{g}'(x) = \frac{\mathrm{g}(x)}{x(1 + \mathrm{g}(x))}.$$

What happens at $x = -\frac{1}{e}$?

3 **An integral.** Use the substitution $x = we^w$ to show that $\displaystyle\int_0^e \mathrm{g}(x)\,\mathrm{d}x = e - 1$.

4 **The tower.** Assuming the tower $T = \sqrt{2}^{\sqrt{2}^{\sqrt{2}^{\cdots}}}$ converges, show that $T = \sqrt{2}^{T}$, and deduce that $T = 2$ or $T = 4$. Which is it? (Compute a few terms of the sequence $t_1 = \sqrt{2}$, $t_{n+1} = \sqrt{2}^{t_n}$, and think about the cobweb diagrams of Section 2.2.)

Further reading

- **Robert Corless et al., "On the Lambert W function"**, *Advances in Computational Mathematics* 5 (1996). The paper that named the function. The first sections are readable with this section behind you.
- **Brian Hayes, "Why W?"**, *American Scientist* 93 (2005). A lively essay on the function's history and uses. Free online.
- **Brian Conrad, "Impossibility theorems for elementary integration"**. Lecture notes, available online, explaining Liouville's theory. Advanced.

The full guided solution, and what the question leads to at university, is in Section 3.9:
https://beyondstep.keithmcnulty.org/ch03/lambert-w.html